

Bank Loan Case Study

Description:

This project performs Exploratory Data Analysis (EDA) on a loan application dataset to identify the factors that influence a customer's repayment behavior.

The dataset contains information about applicant's demographic details, financial status and loan related attributes. By comparing customers who experienced payment difficulties with those who successfully repaid their loans, the analysis aims to uncover patterns that can help improve credit risk assessment.

Business problem -

The company faces two major risks -

1. Approving a loan to a customer who may default.
2. Rejecting a loan to a customer who is capable of repayment.

The goal is to discover patterns associated with payment difficulties and support better lending decisions.

Approach:

Understanding the dataset -

Firstly, I went through the dataset structure and variable provided. There were three dataset files.

1. *application_data.csv* which contains information of the clients at the time of application.
2. *previous_application.csv* which contains information of client's previous loan data.
3. *columns_description.csv* which contains the meaning of the variables present in the above two datasets.

I identified the numerical and categorical features such as the customer demographics (age, gender, etc.), financial information, etc. And then checked the target variable indicating the payment difficulty. Here, Target 0 means no payment difficulty and Target 1 means payment difficulty.

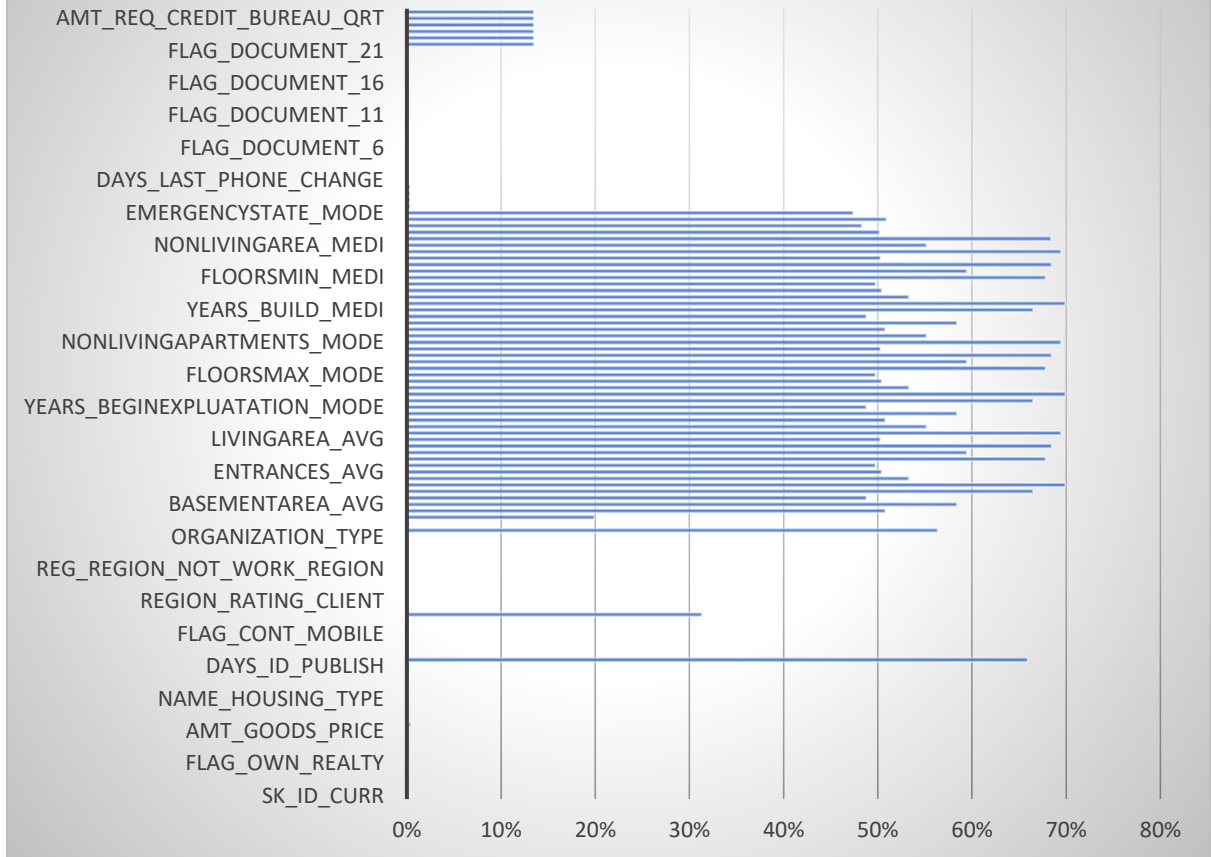
Data Cleaning -

For both the dataset, I checked the Null value percentage for all the columns and dropped the columns having null value percentage more than 30% along with columns which were not relevant for the analysis.

I also changed few numerical data columns like Days_Birth into Year value and removed the negative signs.

Dataset *application_data*

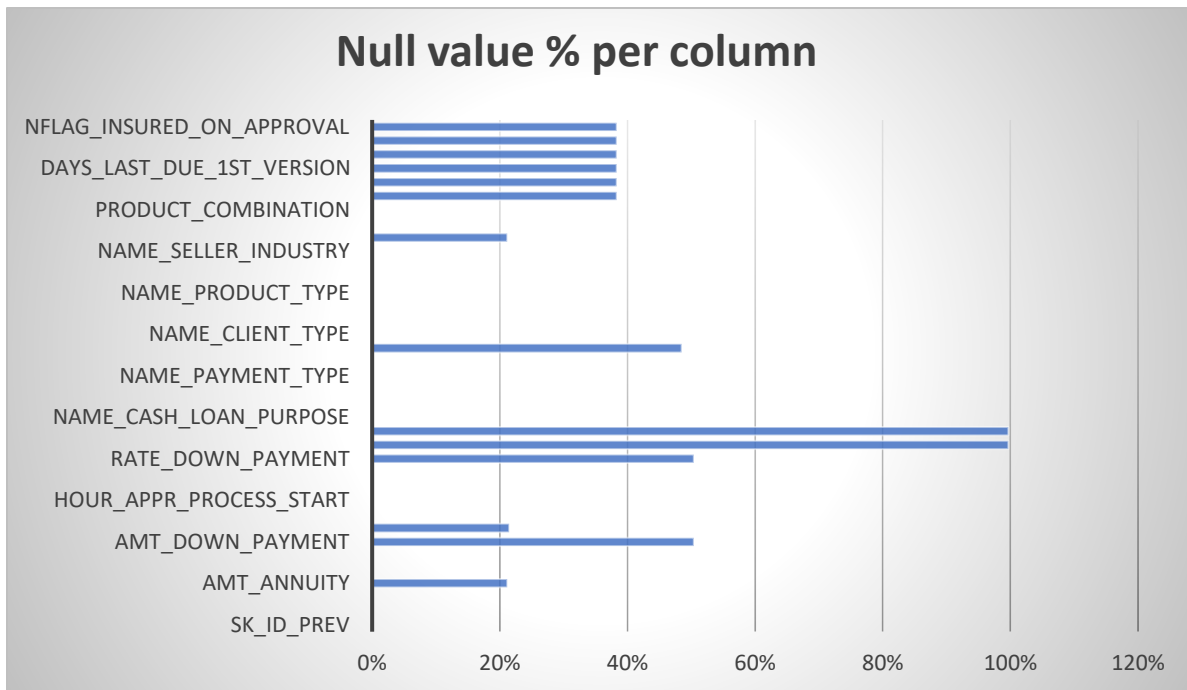
Null value % per column



Irrelevant Columns

FLAG_MOBIL
FLAG_EMP_PHONE
FLAG_WORK_PHONE
FLAG_CONT_MOBILE
FLAG_PHONE
FLAG_EMAIL
FLAG_DOCUMENT_2
FLAG_DOCUMENT_3
FLAG_DOCUMENT_4
FLAG_DOCUMENT_5
FLAG_DOCUMENT_6
FLAG_DOCUMENT_7
FLAG_DOCUMENT_8
FLAG_DOCUMENT_9
FLAG_DOCUMENT_10
FLAG_DOCUMENT_11
FLAG_DOCUMENT_12
FLAG_DOCUMENT_13
FLAG_DOCUMENT_14
FLAG_DOCUMENT_15
FLAG_DOCUMENT_16
FLAG_DOCUMENT_17
FLAG_DOCUMENT_18
FLAG_DOCUMENT_19
FLAG_DOCUMENT_20
FLAG_DOCUMENT_21

Dataset *previous_application*



Irrelevant Columns
FLAG_LAST_APPL_PER_CONTRACT
NFLAG_LAST_APPL_IN_DAY

Identify missing data and deal with it appropriately -

After the data cleaning process and removing the columns with null value percentage more than 30% along with columns which were not relevant for the analysis, there were still few columns with blank cells.

For them, I used median/mode depending on the data type.

Column Names with Null Value <=30% and blank cells	Value Count	Median/Mode
AMT_ANNUITY	49998	24939
AMT_GOODS_PRICE	49998	450000
NAME_TYPE_SUITE	49998	Unaccompanied
CNT_FAM_MEMBERS	49998	2
EXT_SOURCE_2	49873	0.565585366
EXT_SOURCE_3	40055	0.53527625
OBS_30_CNT_SOCIAL_CIRCLE	49831	0
DEF_30_CNT_SOCIAL_CIRCLE	49831	0
OBS_60_CNT_SOCIAL_CIRCLE	49831	0
DEF_60_CNT_SOCIAL_CIRCLE	49831	0
YEARS_LAST_PHONE_CHANGE	49998	2
AMT_REQ_CREDIT_BUREAU_HOUR	43265	0
AMT_REQ_CREDIT_BUREAU_DAY	43265	0
AMT_REQ_CREDIT_BUREAU_WEEK	43265	0
AMT_REQ_CREDIT_BUREAU_MON	43265	0
AMT_REQ_CREDIT_BUREAU_QRT	43265	0
AMT_REQ_CREDIT_BUREAU_YEAR	43265	1

Dataset *application_data*

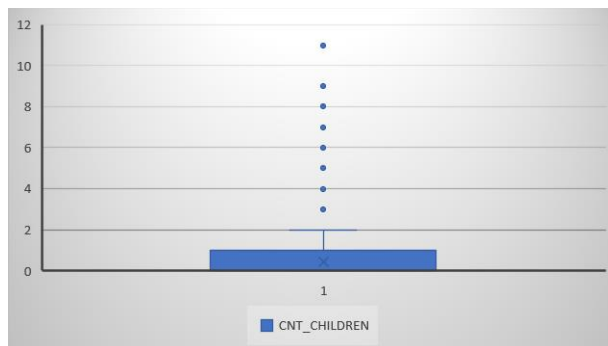
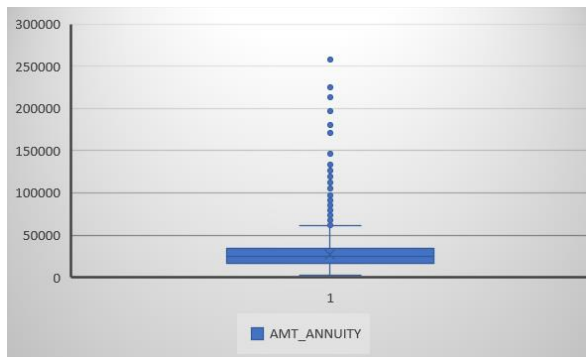
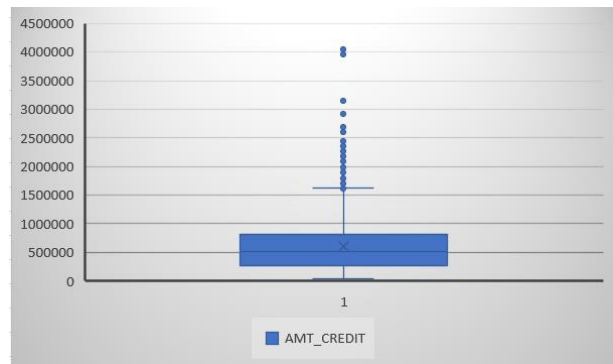
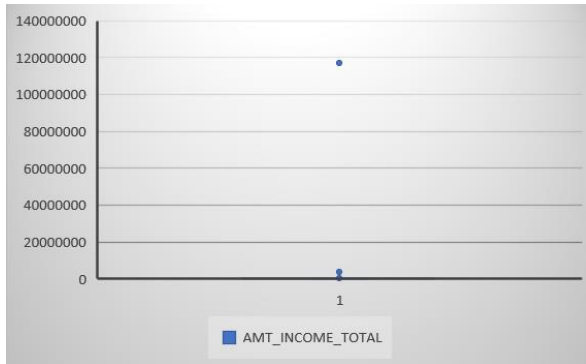
Column Names with Null Value <=30% and blank cells	Value Count	Median/Mode
AMT_ANNUITY	39407	10879.92
AMT_GOODS_PRICE	39255	104017.5
CNT_PAYMENT	39407	12
PRODUCT_COMBINATION	49991	POS mobile with interest

Dataset *previous_application*

Identify Outliers in the dataset -

Here using the Excel functionals, I calculated the Quartiles, IQR, Lower bound and Upper Bound to identify the Outliers.

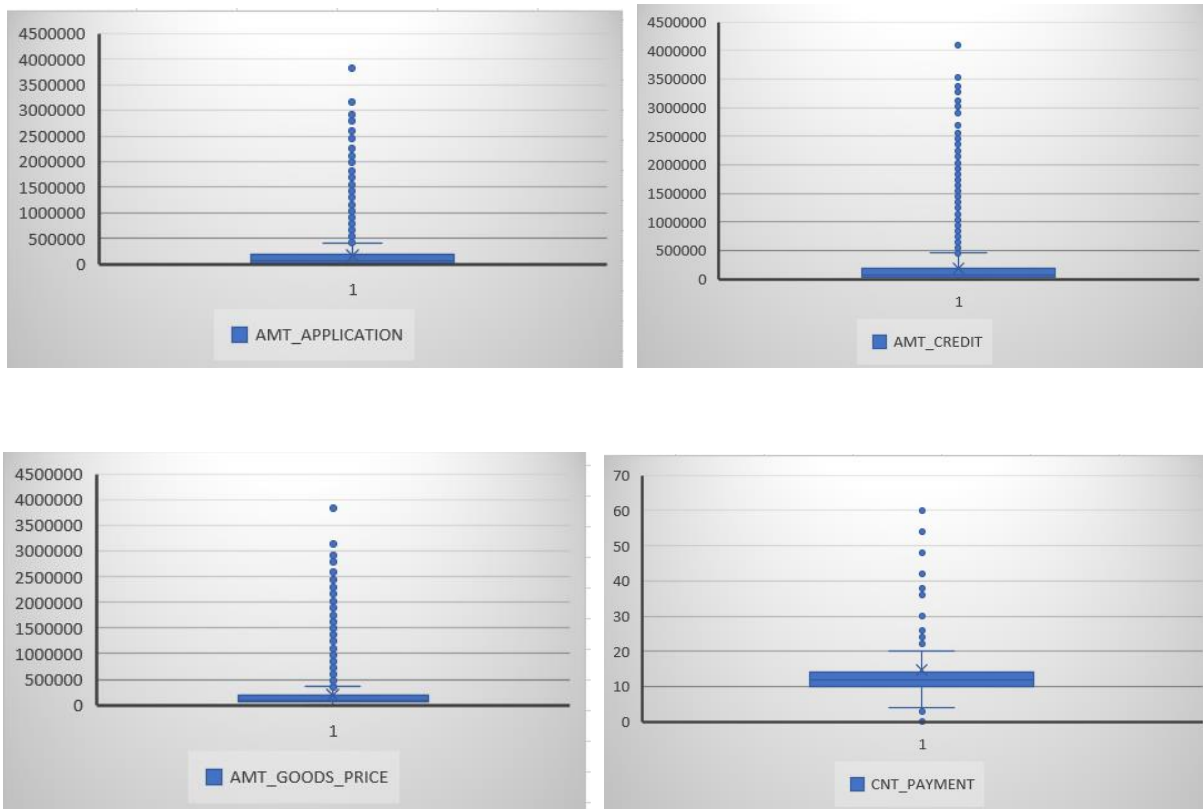
Dataset *application_data*



- AMT_INCOME_TOTAL: A few applicants have much higher incomes than most people in the dataset.
- AMT_CREDIT: Some cases show unusually high loan amounts compared to the general population.

- AMT_ANNUITY: A small number of loans have much higher monthly repayment amounts than usual.
- AMT_GOODS_PRICE: Some loans are linked to very high-value goods compared to most applications.
- CNT_CHILDREN: A few households have unusually high number of children compared to most applicants.
- YEARS_BIRTH: No outliers found, meaning age values are well distributed within a normal range.

Dataset *previous_application*



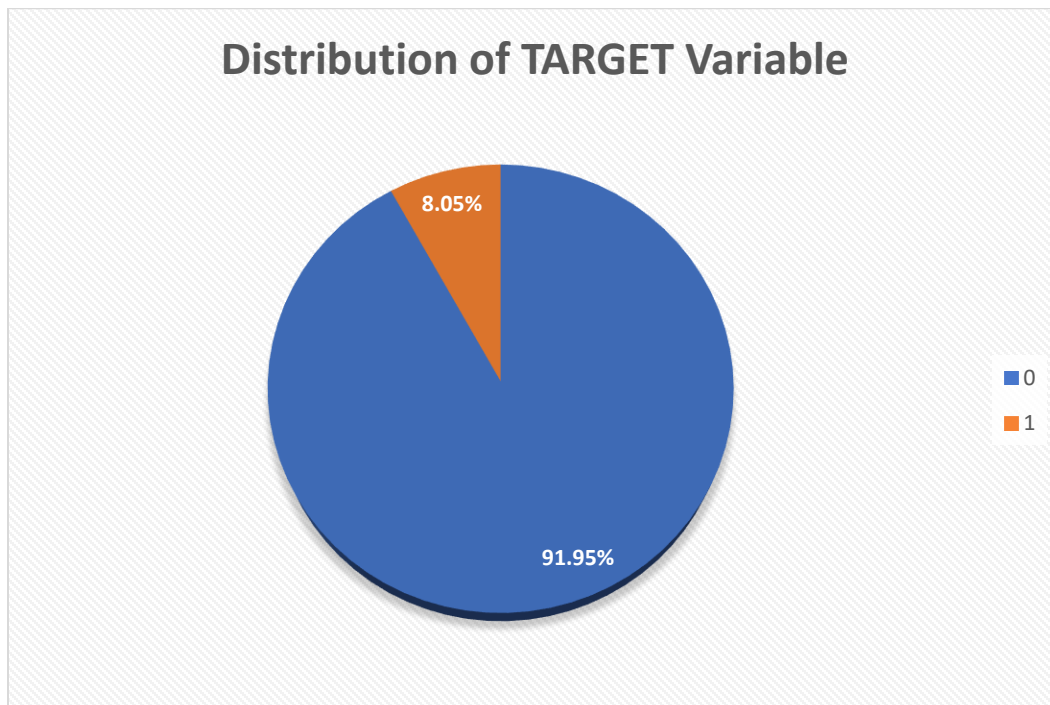
- AMT_APPLICATION: A small number of applications have much higher loan amounts than most others.
- AMT_CREDIT: A few cases show unusually high loan amounts compared to the general pattern.
- AMT_GOODS_PRICE: Some loans are linked to very high-value goods compared to most applications.
- CNT_PAYMENT: A number of cases have unusually long repayment periods compared to normal loans.

Analyze Data Imbalance -

Here the Data Imbalance is shown between clients who are making payment on time (Target 0) and clients facing difficulties with payment (Target 1).

Dataset *application_data*

Target	Meaning	Count	Percentage
0	No Payment Difficulty	45973	91.95%
1	Payment Difficulty	4026	8.05%
Total		49999	100%



And according to the output the dataset contains significantly more non-defaulters than defaulters which indicates data imbalance.

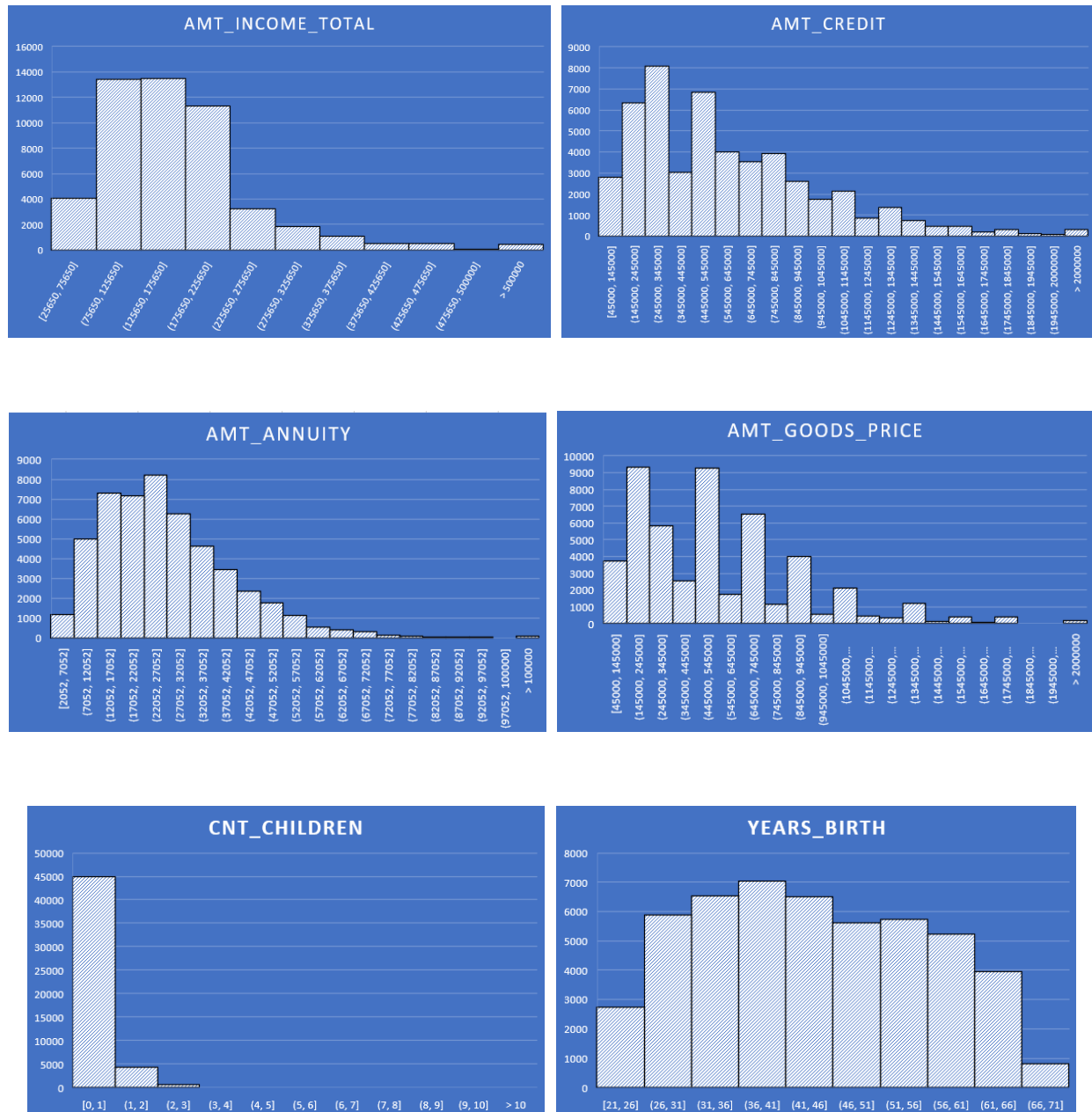
Perform Univariate, Segmented Univariate, and Bivariate Analysis -

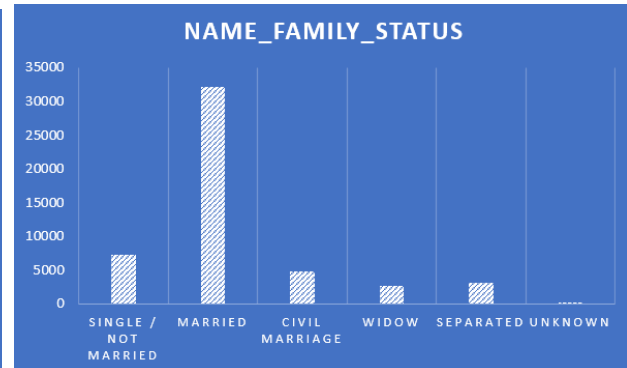
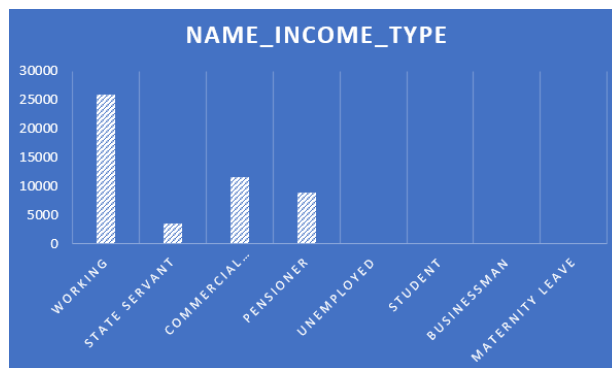
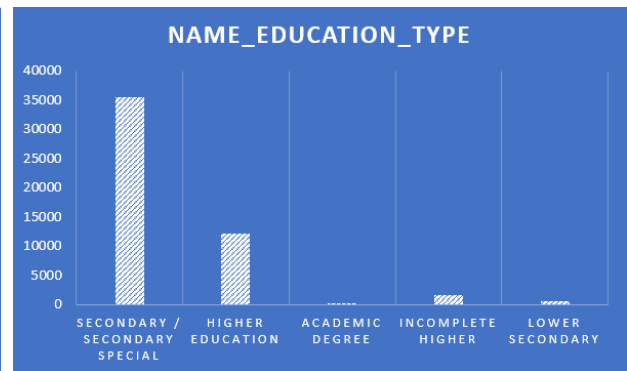
Here, the main objective of EDA was to understand how client characteristics influence loan repayment behavior.

Univariate Analysis -

This analysis studies one variable at a time, so individual variables were analyzed separately to understand their distribution. I analyzed the numerical and the categorical variables.

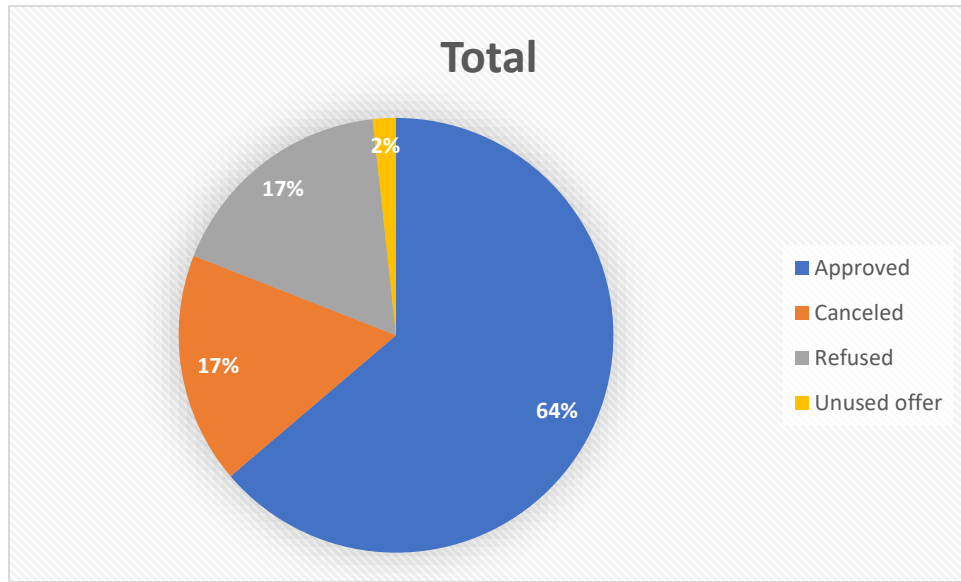
Dataset *application_data*





- AMT_INCOME_TOTAL - Most applicants earn a moderate income, while only a few have very high earnings.
- AMT_CREDIT - Most customers apply for average-sized loans, with large loan requests being less common.
- AMT_ANNUITY - Most borrowers have affordable monthly instalment amounts.
- AMT_GOODS_PRICE - Customers generally purchase moderately priced goods rather than expensive assets.
- CNT_CHILDREN - Most applicants have small families, with up to two children.
- YEARS_BIRTH - The majority of applicants are working-age adults.
- CODE_GENDER - More women apply for loans than men.
- NAME_EDUCATION_TYPE - Secondary education is the most common qualification among applicants.
- NAME_INCOME_TYPE - Most loan applications come from working professionals.
- NAME_FAMILY_STATUS - Married individuals make up the largest group of applicants.

Dataset *previous_application*

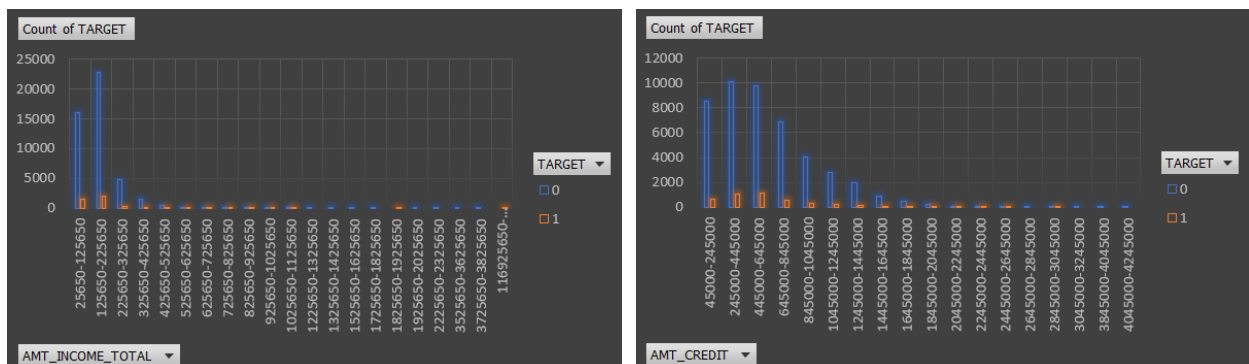


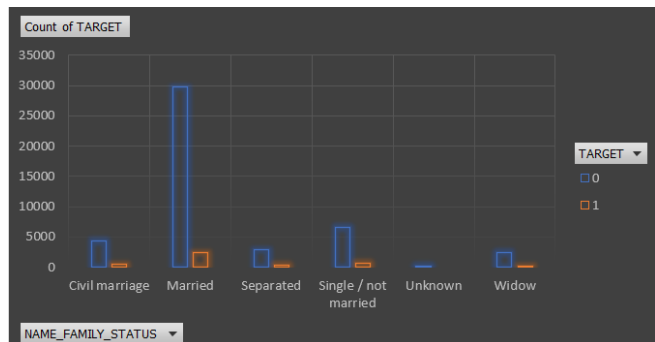
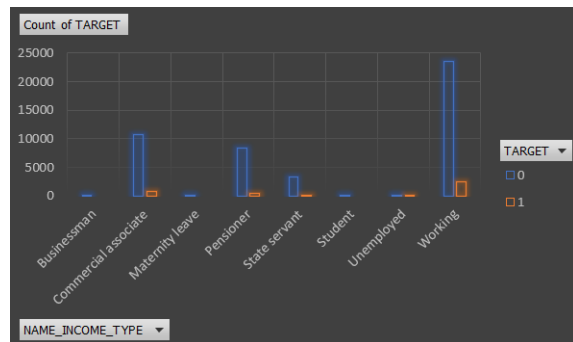
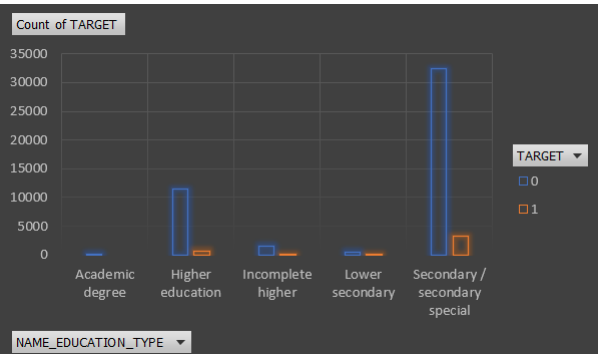
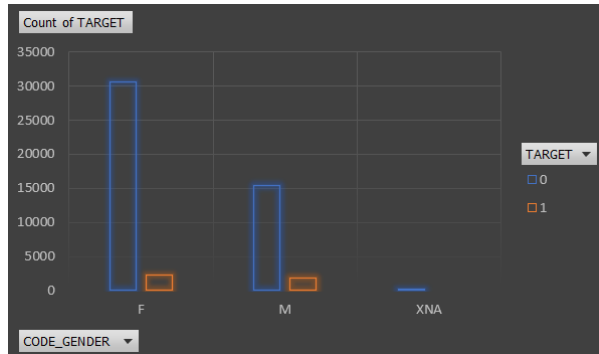
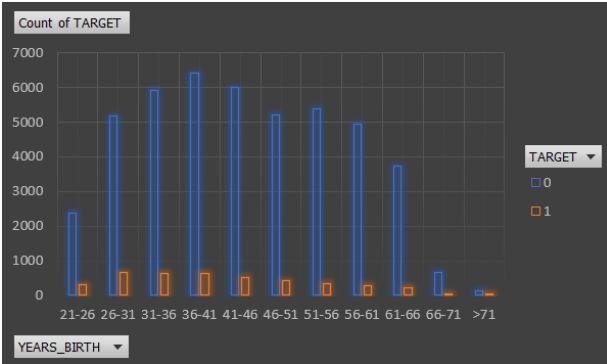
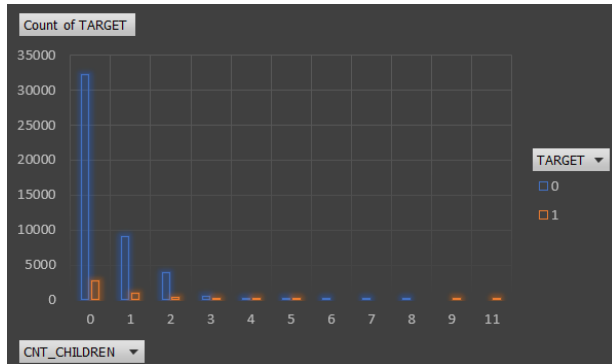
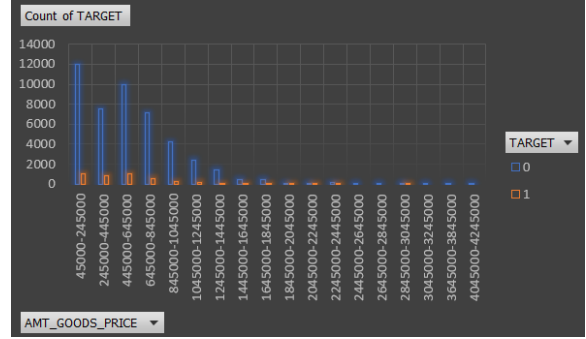
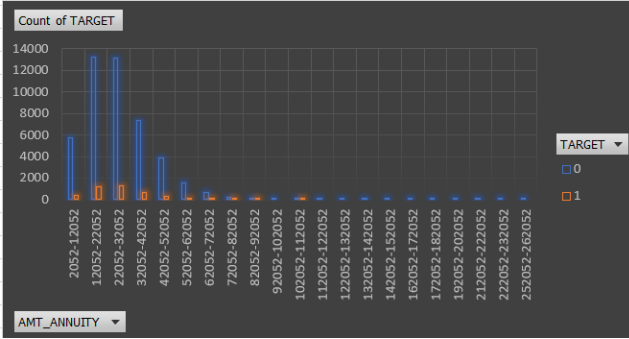
Here, most previous applications are approved, while a smaller portion are refused or cancelled and very few are unused offers.

Segmented Univariate Analysis -

This analysis compares variable distribution across 'Target' categories.

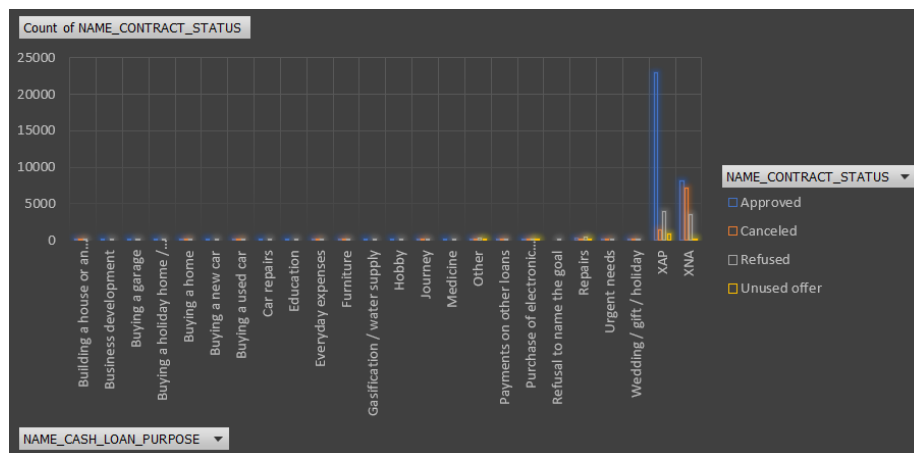
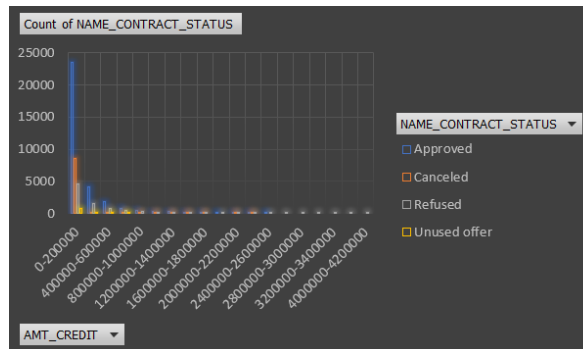
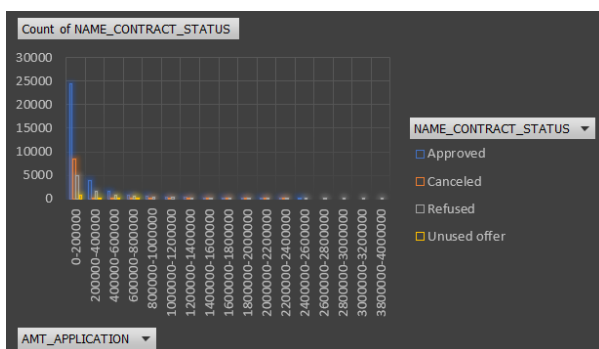
Dataset *application_data*

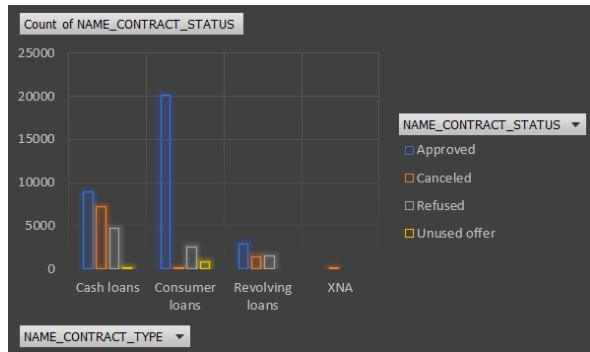
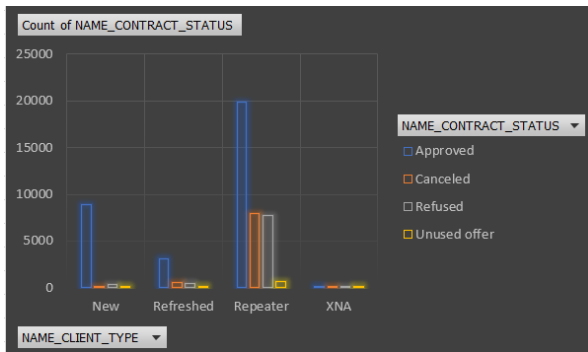




- AMT_INCOME_TOTAL vs TARGET - Payment difficulties are more common among lower-income applicants.
- AMT_CREDIT vs TARGET - Defaults occur across all loan amounts, not just larger loans.
- AMT_ANNUITY vs TARGET - Monthly instalments are similar for both groups.
- AMT_GOODS_PRICE vs TARGET - Payment difficulties are more common among customers purchasing lower to mid-priced goods.
- CNT_CHILDREN vs TARGET - Applicants with children show slightly higher default rates.
- YEARS_BIRTH vs TARGET - Younger applicants are more likely to face payment difficulties.
- CODE_GENDER vs TARGET - More women apply for loans, though repayment patterns differ slightly by gender.
- NAME_EDUCATION_TYPE vs TARGET - Higher education is associated with lower default rates.
- NAME_INCOME_TYPE vs TARGET - Default rates vary across different income sources.
- NAME_FAMILY_STATUS vs TARGET - Married applicants make up the largest share of both groups.

Dataset *previous_application*



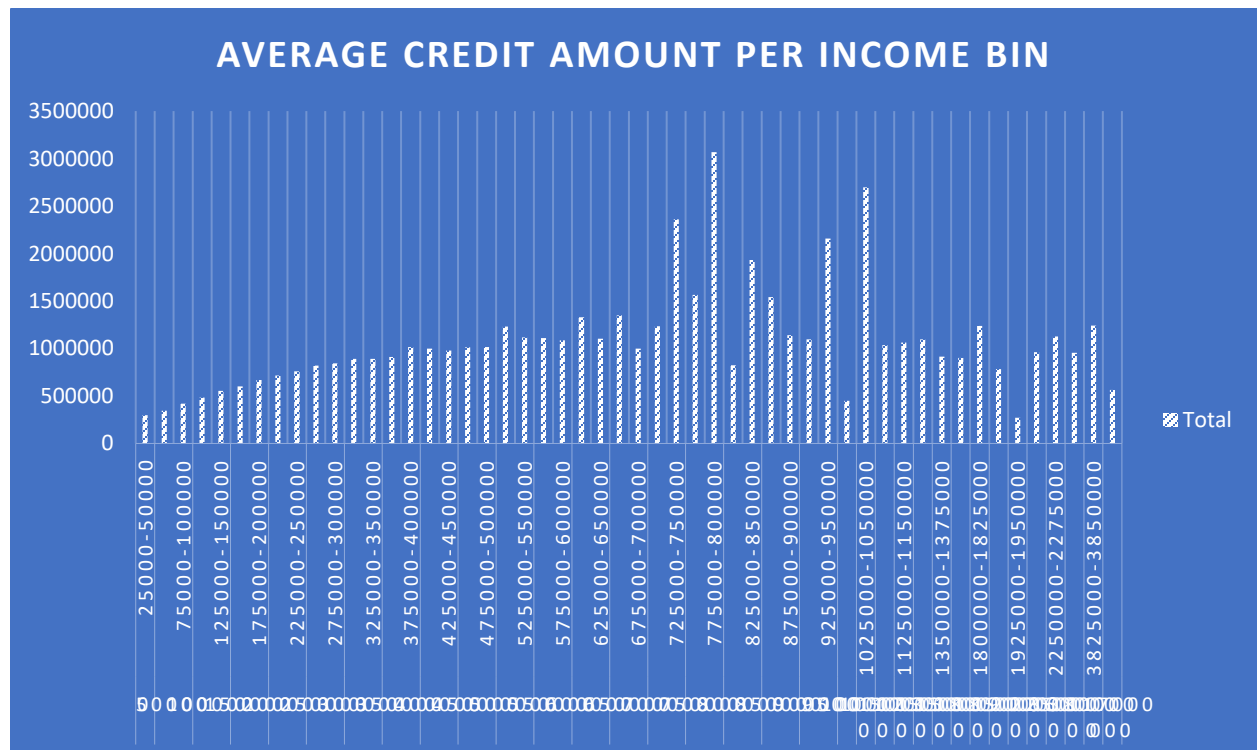


- AMT_APPLICATION - Most loans are in the low amount range, and higher loan amounts are rare.
- AMT_CREDIT - Small credit loans make up the majority of applications.
- NAME_CASH_LOAN_PURPOSE - Repairs and XNA are the most common purposes, but they also have many refusals.
- NAME_CLIENT_TYPE - Repeat customers make up most of the applications across all statuses.
- NAME_CONTRACT_TYPE - Consumer loans are most often approved, while cash loans have more mixed outcomes.

Bivariate Analysis -

This analysis studies relationship between two variables to identify possible risk indicators.

Dataset *application_data*



Here, as income increases the average credit amount generally increases, indicating that higher-income applicants tend to qualify for larger loans.

Identify Top Correlation for Different Scenarios -

I performed the Correlation analysis to identify variables that have the strongest relationship with loan repayment behavior using Excel function CORREL.

Client with payment difficulty -

TARGET = 1	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	CNT_CHILDREN	YEARS_BIRTH	YEARS_EMPLOYED	REGION_POPULATION_RELATIVE	REGION_RATING_CLIENT
AMT_INCOME_TOTAL	1								
AMT_CREDIT	0.015	1							
AMT_ANNUITY	0.018	0.750	1						
AMT_GOODS_PRICE	0.013	0.982	0.750	1					
CNT_CHILDREN	0.010	0.008	0.029	-0.001	1				
YEARS_BIRTH	-0.009	0.143	0.009	0.141	-0.250	1			
YEARS_EMPLOYED	-0.012	0.019	-0.078	0.023	-0.190	0.588	1		
REGION_POPULATION_RELATIVE	-0.006	0.068	0.073	0.077	-0.020	0.016	0.008	1	
REGION_RATING_CLIENT	-0.013	-0.045	-0.062	-0.051	0.056	-0.045	-0.009	-0.430	1

Clients without payment difficulty –

TARGET = 0	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	CNT_CHILDREN	YEARS_BIRTH	YEARS_EMPLOYED	REGION_POPULATION_RELATIVE	REGION_RATING_CLIENT
AMT_INCOME_TOTAL	1								
AMT_CREDIT	0.378	1							
AMT_ANNUITY	0.451	0.771	1						
AMT_GOODS_PRICE	0.385	0.987	0.776	1					
CNT_CHILDREN	0.036	0.006	0.026	0.002	1				
YEARS_BIRTH	-0.074	0.051	-0.010	0.049	-0.336	1			
YEARS_EMPLOYED	-0.162	-0.075	-0.111	-0.072	-0.246	0.623	1		
REGION_POPULATION_RELATIVE	0.182	0.096	0.117	0.099	-0.025	0.030	-0.007	1	
REGION_RATING_CLIENT	-0.205	-0.103	-0.130	-0.105	0.021	-0.009	0.041	-0.539	1

The strongest relationships in both groups are between AMT_CREDIT, AMT_GOODS_PRICE and AMT_ANNUITY showing that these variables naturally move together. Since the correlation patterns are very similar for both defaulters and non-defaulters, loan default is likely influenced by a combination of customer characteristics and financial circumstances rather than any single factor.

Tech-Stack Used:

Microsoft Excel Worksheet (2019) - Used for Data cleaning and analysis

Excel Pivot tables - Used for Data summarization

Excel Charts - Used for Data visualization

Excel statistical functions - Used for descriptive and correlation analysis

Microsoft word/PDF - Used for documentation making report for the output and insights.

Insights:

After all the analysis, the dataset shows that most applicants are middle-income working adults with small families, applying for moderate loan amounts and affordable instalments, while only a small portion have very high income or large loan values. The data is heavily imbalanced with far more non-defaulters than defaulters, which suggests most customers repay loans successfully.

Previous application analysis also shows that most loans are approved, especially for lower loan amounts, while higher loan values and long repayment terms appear less frequently and carry more risk. Customer segmentation indicates that repeat applicants form the largest group, and consumer loans are the most commonly approved loan type, whereas cash loans show more mixed outcomes.

From a risk perspective, payment difficulties are more likely among younger applicants, lower-income groups, and those with children, while higher education levels are associated with better repayment behavior. Correlation analysis shows that income is positively related to credit amount, and key financial variables like credit, annuity, and goods price move together strongly in both defaulters and non-defaulters, suggesting that loan default is not driven by a single factor but by a combination of financial capacity and borrower profile.

Results:

This project provided a comprehensive understanding of applying Exploratory Data Analysis to understand customer repayment behavior and identify factors with loan default.

Through missing value treatment, outlier detection, imbalance analysis and exploratory analysis, valuable insights were provided regarding customer risk profiles. These can support more informed lending decisions, improve credit risk management and reduce potential financial losses.

This project also helped improve my risk analytics skills while helping me understand the application and importance of EDA more clearly.

Drive link:

<https://drive.google.com/drive/folders/17AaNv5KLNEWgjVr-QzkNq0Aneb5Dd0V4?usp=sharing>

[Click here for Excel Sheet file Dataset *application data*](#)

[Click here for Excel Sheet file Dataset *previous application*](#)

[Click here for Video Presentation](#)